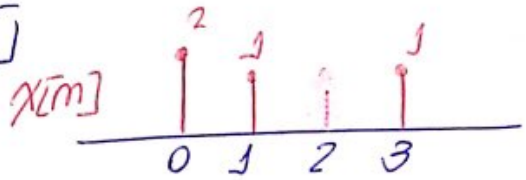
$m] = 2\delta[m] + \delta[m-1] + \delta[m-3]$



Calculo la DFT de 5 puntos de $x[m]$

•) DFT inversa de (5 puntos) $Y[k] = |X[k]|^2$

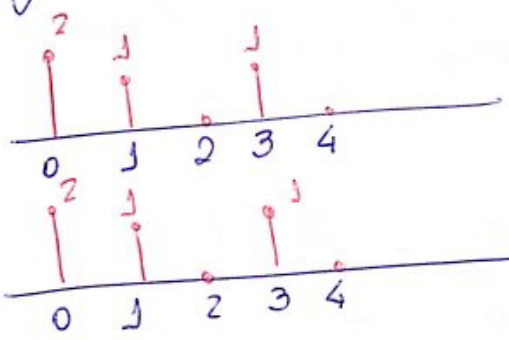
a) Determinar $g[m]$; $m = 0,1,2,3,4$

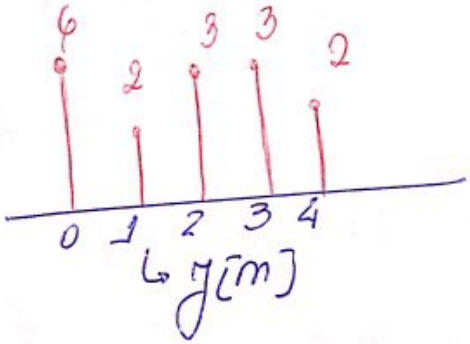
$$Y[k] = |X[k]|^2 = X[k] \cdot X^*[k]$$

↳ $g[m] = x[m] \odot x^*[-m]$

Pero $x[m]$ es real $\Rightarrow x^*[m] = x[m]$

$g[m] = x[m] \odot x[-m]$





↳ $g[m]$

$$\Rightarrow g[m] = 6\delta[m] + 2\delta[m-1] + 3\delta[m-2] + 3\delta[m-3] + 2\delta[m-4]$$

b) $g[m] = x[m] * x[m]$

$x[m]$ tiene longitud 4

$x[m] * x[m]$ tiene longitud $N = 4+4-1 = 7$

$\rightarrow \boxed{N = 7}$

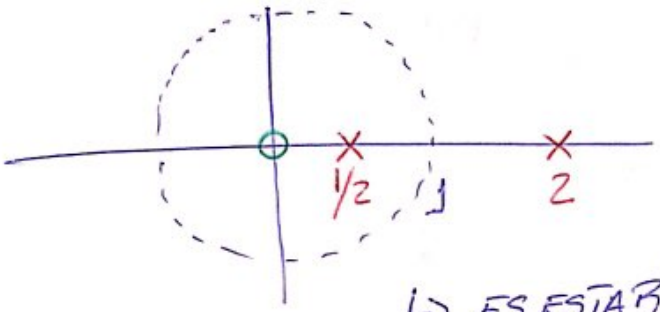
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2) $r[m] = \frac{4}{3}\left(\frac{1}{2}\right)^m u[m] + \frac{4}{3} 2^m u[-m-1]$

a) $R(z) = \frac{4}{3}\left\{ \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{-1}{1-2z^{-1}} \right\}$

$|z| > 1/2 \qquad |z| < 2.$

Roc: $1/2 < |z| < 2.$



↳ ES ESTABLE PERO NO ES CAUSAL

b) $r[m] = h[m] * h[-m]$

↳ $R(z) = H(z)\,H(z^{-1}) = \frac{4}{3}\left\{ \frac{z}{z-1/2} - \frac{z}{z-2} \right\}$

$R(z) = \frac{4}{3} \frac{z(z-2) - z(z-1/2)}{(z-1/2)(z-2)} = \frac{4}{3} \frac{z^2 - 2z - z^2 + 1/2\,z}{(z-1/2)(z-2)}$

$R(z) = \frac{4}{3} \frac{-3/2\,z}{(z-1/2)(z-2)} = \frac{-2z}{(z-1/2)(z-2)}$

PROPONGO $H(z) = \frac{z}{z-1/2}$, $h[m]$ DE FASE MÍNIMA

↳ POLOS Y CEROS DENTRO DE LA CIRCUNFERENCIA UNIDAD

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$$H(z^{-1}) = \frac{z^{-1}}{z^{-1} - 1/2} = \frac{1}{1 - z/2} = \frac{2}{2 - z}$$

$$H(z^{-1}) = \frac{-2}{z - 2}$$

$$\Rightarrow H(z)\,H(z^{-1}) = \frac{z}{z - 1/2} \cdot \frac{-2}{z - 2}$$

$$H(z)\,H(z^{-1}) = \frac{-2z}{(z - 1/2)(z - 2)} = R(z)$$

3) $$H_3(j\omega) = \begin{cases} T; & |\omega| < \pi/T \\ 0; & \text{otro caso.} \end{cases}$$

$$h_2(t): \quad \frac{d^2 y_c(t)}{dt^2} + \alpha \frac{dy_c(t)}{dt} + \beta y_c(t) = x_c(t)$$

TRANSFORMANDO:

$$(j\omega)^2 Y_c(j\omega) + \alpha j\omega Y_c(j\omega) + \beta Y_c(j\omega) = X_c(j\omega)$$

$$-\omega^2 Y_c(j\omega) + j\omega\alpha Y_c(j\omega) + \beta Y_c(j\omega) = X_c(j\omega)$$

$$Y_c(j\omega)\{-\omega^2 + j\omega\alpha + \beta\} = X_c(j\omega)$$

$$\Rightarrow H_c(j\omega) = \frac{Y_c(j\omega)}{X_c(j\omega)} = \frac{1}{-\omega^2 + j\omega\alpha + \beta}$$

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$$Hd(e^{j\Omega}) = H_1\left(j\frac{\Omega}{T}\right) H_2\left(j\frac{\Omega}{T}\right)$$

$$H_1\left(j\frac{\Omega}{T}\right) = \begin{cases} T\,; & |\Omega| < \pi \\ 0\,; & \text{otro caso.} \end{cases}$$

$$\Rightarrow Hd(e^{j\Omega}) = T\,\frac{1}{\left(\frac{\Omega}{T}\right)^2 + j\frac{\Omega}{T}\alpha + \beta} \qquad |\Omega| < \pi$$

$$\boxed{Hd(e^{j\Omega}) = \frac{T}{\left(\Omega/T\right)^2 + j\frac{\alpha}{T}\Omega + \beta}.}$$

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